Table 1: Success rate on the factored critic×actor seam: mean over each run's last 3 eval points, ± sd over 3 seeds. **BC** is the `bcflow` control (the Flow actor with `q_coeff`=0, so the critic's value term is removed). Only FB's BC control is available in this source table; because the actor conditions on the critic's conditioning vector (FB's learned z for FB, the goal for the goal-conditioned critics), that row is not the matched control for the GCIQL, biGCIQL and CRL blocks.

		ManiSkill				Metaworld			
		`ms-pick-cube`		`ms-push-cube`		`mw-pick-place`		`mw-push`	
Critic	Actor	expert	play-noise	expert	play-noise	expert	play-noise	expert	play-noise
FB	BC	0.19 ± 0.01	0.57 ± 0.06	0.09 ± 0.05	0.82 ± 0.04	1.00 ± 0.00	0.88 ± 0.03	1.00 ± 0.00	0.93 ± 0.04
	Flow	0.21 ± 0.08	0.69 ± 0.08	0.04 ± 0.04	0.78 ± 0.13	1.00 ± 0.00	0.86 ± 0.03	1.00 ± 0.00	0.96 ± 0.02
	DDPG	0.20 ± 0.03	0.20 ± 0.04	0.10 ± 0.10	0.58 ± 0.08	1.00 ± 0.00	0.91 ± 0.02	1.00 ± 0.00	0.88 ± 0.04
GCIQL	Flow	0.08 ± 0.02	0.14 ± 0.03	0.01 ± 0.01	0.31 ± 0.16	0.38 ± 0.04	0.88 ± 0.06	0.18 ± 0.02	0.94 ± 0.03
	DDPG	0.05 ± 0.00	0.06 ± 0.01	0.00 ± 0.00	0.13 ± 0.05	0.00 ± 0.00	0.58 ± 0.01	0.00 ± 0.00	0.80 ± 0.03
biGCIQL	Flow	0.08 ± 0.03	0.08 ± 0.03	0.00 ± 0.00	0.19 ± 0.06	0.34 ± 0.13	0.85 ± 0.10	0.37 ± 0.24	0.90 ± 0.02
	DDPG	0.06 ± 0.01	0.05 ± 0.00	0.00 ± 0.00	0.16 ± 0.13	0.01 ± 0.01	0.39 ± 0.15	0.10 ± 0.08	0.64 ± 0.05
CRL	Flow	0.05 ± 0.00	0.10 ± 0.04	0.01 ± 0.01	0.43 ± 0.14	0.79 ± 0.07	0.53 ± 0.39	0.64 ± 0.55	0.22 ± 0.10
	DDPG	0.05 ± 0.00	0.06 ± 0.01	0.01 ± 0.01	0.28 ± 0.04	0.40 ± 0.30	0.07 ± 0.04	0.73 ± 0.09	0.10 ± 0.04

Table 2: Pre-registered kill-test arms (phenomenon.tex), launched 2026-08-03. *play-noise-hard* = scripted-oracle mixture with `expert_frac`=0, $\varepsilon_{\max} = 0.5$, $\sigma_{\max} = 0.3$ (reopens headroom, Ph. I only); *play-resample* = OGBench-style play: 200 trajectories × 500 steps, target re-drawn from the env goal space on every success (no dwell) and every 60 steps (equalizes E7: horizon, dwell, goal diversity, kernel range; eval-side goal channel untouched). Cube arms test the Ph. V mechanism. All cells 3 seeds; PENDING = submitted, awaiting results.

Setting	Critic	Actor	`mw-push`	`mw-pick-place`
play-noise-hard (Pred. headroom)	biGCIQL	BC	0.600 ± 0.050	0.533 ± 0.058
	biGCIQL	Flow	$\mathbf{0.867 \pm 0.076}$	0.583 ± 0.058
	GCIQL	BC	0.667 ± 0.029	0.517 ± 0.076
	GCIQL	Flow	0.717 ± 0.076	0.483 ± 0.058
play-resample (Pred. kernel(b))	biGCIQL	BC	0.967 ± 0.029	0.950 ± 0.000
	biGCIQL	Flow	0.983 ± 0.029	0.983 ± 0.029
	GCIQL	BC	0.967 ± 0.029	0.983 ± 0.029
	GCIQL	Flow	0.950 ± 0.000	0.983 ± 0.029
play-resample-hard ($\varepsilon_{\max}$=0.5, $\sigma_{\max}$=0.3)	biGCIQL	BC	0.683 ± 0.126	0.550 ± 0.050
	biGCIQL	Flow	0.700 ± 0.087	$\mathbf{0.667 \pm 0.029}$
	GCIQL	BC	0.600 ± 0.100	0.617 ± 0.115
	GCIQL	Flow	0.650 ± 0.050	0.617 ± 0.076

Cube arm (500k, 3 seeds)	Tests	Success
GCIQL + $\kappa \tanh(\cdot/\kappa)$, κ=75 (`gciql-qb75`)	Ph. V(a) boundedness	0.760 ± 0.037
biGCIQL d=50, no ψ-LayerNorm (`biql-d50-nopsiln`)	Ph. V(b) norm concentration	0.909 ± 0.032
biGCIQL d=15	$d^\dagger$ bracket	0.837 ± 0.027

Datasets

Every dataset above is one packed `npz` per task at `<root>/data/<family>/<task>.npz`, the eight unified fields stacked $[N, T, \cdot]$ (`observation`, `action`, `next_observation`, `achieved_goal`, `reward`, `terminal`, `timeout`, `success`). `env/unified.py::load_packed` flattens to $N \cdot T$ transitions and *synthesizes* the `terminals` that mark trajectory boundaries at each episode's last step: the env's own `terminal` flag never fires, since every collection here is fixed-horizon. Two schema facts matter for reading the grid. (i) `achieved_goal` is a zeros placeholder in all six datasets and is never read. (ii) The eval goal for the goal-conditioned critics does not come from the data's goal field either — it is the single highest-reward `next_observation` in the dataset (`env/unified.py:167`), one fixed frame reused for every eval episode. The *play-resample* arm deliberately leaves this eval-side channel unrepaired.

All statistics in Table 3 were recomputed from the packed files, not copied from the collection logs. *ep-succ* is the fraction of episodes in which `success` ever fires; *dwell* is the fraction of *all* steps spent at the goal — the quantity that separates "reached it" from "sat on it".

Table 3: Provenance and measured content of every dataset used above. The exploration mixture is drawn once per episode (per *segment* for the resample regimes): a coin f_{exp} decides whether the episode stays clean, otherwise $\varepsilon \sim U(0, \varepsilon_{\max})$ and $\sigma \sim U(0, \sigma_{\max})$ give $a = \text{clip}(\pi(s) + \sigma\mathcal{N}(0, I))$, replaced wholesale by $U(-1, 1)$ with probability ε.

Regime	Task	Collector / policy	$N \times T$	obs/act	$\varepsilon_{\max}/\sigma_{\max}/f_{\text{exp}}$	ep-succ	dwell	reloc/traj
`expert`	`ms-pick-cube`	newt TD-MPC2 ckpt, planner on	1000×25	42/4	0/0/1	0.996	0.746	–
	`ms-push-cube`	newt TD-MPC2 ckpt, planner on	1000×25	35/7	0/0/1	0.999	0.858	–
	`mw-pick-place`	MW scripted oracle	1000×100	39/4	0/0/1	1.000	0.749	–
	`mw-push`	MW scripted oracle	1000×100	39/4	0/0/1	0.999	0.706	–
`play-noise`	`ms-pick-cube`	TD-MPC2, `mpc=false`	1000×25	42/4	0.25/0.15/0.3	0.772	0.272	–
	`ms-push-cube`	TD-MPC2, `mpc=false`	1000×25	35/7	0.25/0.15/0.3	0.955	0.691	–
	`mw-pick-place`	MW scripted oracle	1000×100	39/4	0.3/0.2/0.3	0.860	0.531	–
	`mw-push`	MW scripted oracle	1000×100	39/4	0.3/0.2/0.3	0.820	0.484	–
`play-noise-hard`	`mw-pick-place`	as `play-noise`, hotter mixture	1000×100	39/4	0.5/0.3/0	0.544	0.245	–
	`mw-push`	as `play-noise`, hotter mixture	1000×100	39/4	0.5/0.3/0	0.500	0.209	–
`play-resample`	`mw-pick-place`	oracle + target re-draw	200×500	39/4	0.1/0.1/0	0.960	0.095	47.5
	`mw-push`	oracle + target re-draw	200×500	39/4	0.1/0.1/0	0.965	0.062	31.0
`play-resample-hard`	`mw-pick-place`	oracle + target re-draw	200×500	39/4	0.5/0.3/0	0.655	0.007	3.43
	`mw-push`	oracle + target re-draw	200×500	39/4	0.5/0.3/0	0.710	0.010	4.91
`cube-single`	`cube-single-play-v0`	OGBench (external)	stock OGBench play data; eval = mean success over fixed `task_infos`					

`expert`. ManiSkill comes from the published Factor-FB-Offline pack: rollouts of the pretrained newt TD-MPC2 checkpoints (HF `nicklashansen/newt`, snapshot `7eef11e`) with the planner on and zero exploration, 50 parallel envs, failures kept (`success_only=false`); newt's multitask obs/action padding is stripped back to the raw env dims. That pack has no `mw-pick-place`/`mw-push`, and its play counterpart is unreachable, so both Metaworld tasks were regenerated from Metaworld's built-in scripted oracle (`data_collection/collect_mw_standalone.py`, torch-free, no checkpoint) through `env/metaworld_env.py` — action-repeat 2, horizon 100, byte-identical to the eval env. This regime is a saturated single-mode oracle corridor: BC already solves it, which is why the FB/BC row reads 1.00 ± 0.00 on both Metaworld expert columns and why no critic can earn its keep there (Ph. I).

`play-noise` and `play-noise-hard`. Same collectors, exploration mixture on. Structurally this is still *one target per episode*, held for the whole horizon, with heavy terminal dwell — the oracle reaches the goal and then sits there for roughly half the episode, so the goal marginal collapses onto a single per-episode point. Only the action distribution differs from `expert`, not the goal distribution. `play-noise-hard` changes nothing but the mixture ($f_{\text{exp}} = 0$, $\varepsilon_{\max} = 0.5$, $\sigma_{\max} = 0.3$); horizon, per-episode target and dwell are left untouched *on purpose*, so the arm isolates Ph. I from Ph. II/III. It drops the BC floor to ≈ 0.5, which is the entire point: the benchmark measures guidance quality again rather than measuring the ceiling.

`play-resample` and `play-resample-hard`. A different *shape* of data, from `collect_mw_play_resample.py`: 200 trajectories of 500 wrapper steps (= 1000 simulator steps, cube-play scale). The target is re-drawn from the env's own `goal_space` and written into `raw._target_pos` — exactly the quantity the observation's goal dims `obs`[36:39] and the scripted oracle read — immediately when `success` fires (no dwell, as in cube play) and every `seg_max` = 60 steps regardless, so no segment gets stuck. `raw.curr_path_length` is zeroed before each wrapper step to defeat Metaworld's `max_path_length` = 500 truncation. Coverage here comes from relocation chaining, not action noise, so the mixture

is deliberately gentle. Each trajectory becomes a chain of ∼30–48 reach-and-move segments, and within-trajectory relabeling now crosses target contexts through transitions the data actually contains (the train-side repair of Ph. II). The regime nonetheless re-saturates: everything scores 0.95–0.98, BC included. Under the hot mixture the chaining breaks — relocations per trajectory collapse by an order of magnitude, because the oracle rarely completes a segment — but `play-resample-hard` is thereby the only arm that is simultaneously un-saturated *and* goal-diverse, which is what makes it informative.

Table 4: What each kill-test dataset actually varies, relative to `play-noise`.

Dataset	Headroom (BC floor)	Goal diversity
`play-noise`	none (BC ≈ 0.9)	none (1 target/ep, dwell ≈ 0.5)
`play-noise-hard`	**reopened** (BC ≈ 0.5)	none
`play-resample`	none (BC ≈ 0.96)	**high** (30–48 relocs, dwell ≈ 0.08)
`play-resample-hard`	**reopened** (BC ≈ 0.65)	moderate (3–5 relocs, dwell ≈ 0.01)